Development of an Expandable Explainable AI Framework for Multimodal Prediction of Diabetes, Cardiovascular, Renal and Liver Diseases

Abstract

The increasing prevalence of chronic diseases such as Diabetes Mellitus (DM), Cardiovascular Disease (CVD), Chronic Kidney Disease (CKD), and Chronic Liver Disease (CLD) has created a growing need for intelligent, reliable, and personalized disease prediction systems. Although machine learning (ML) and artificial intelligence (AI) techniques have demonstrated significant potential in healthcare, many existing disease prediction models are developed independently for a single disease and often function as black-box systems, limiting their scalability, interpretability, and clinical applicability. This research proposes an **Expandable and Explainable Multimodal Artificial Intelligence (E-XMAI) framework** for the early prediction and risk stratification of multiple chronic diseases using heterogeneous patient data.

The proposed framework will integrate demographic, clinical, biochemical, lifestyle, and physiological information through a common data-processing and feature-fusion architecture. Four chronic diseases—DM, CVD, CKD, and CLD—will be investigated using appropriate machine learning and deep learning techniques. A shared learning architecture with disease-specific prediction modules will be developed to identify both common and disease-specific patterns in patient data. Explainable Artificial Intelligence (XAI) techniques, including SHAP, LIME, and counterfactual explanations, will be incorporated to identify the major factors influencing individual predictions and provide interpretable risk assessments.

A key contribution of the proposed research is the **expandability of the framework**. Continual and incremental learning strategies will be investigated to enable the addition of new diseases or new patient data while preserving the knowledge acquired from previously learned diseases. The framework will therefore be evaluated not only in terms of predictive performance but also with respect to explainability, robustness, generalizability, computational efficiency, and catastrophic forgetting during incremental disease learning. Model performance will be assessed using accuracy, precision, recall, F1-score, sensitivity, specificity, AUROC, AUPRC, calibration, and explainability consistency.

The proposed framework aims to provide a scalable and transparent AI-based decision-support approach capable of identifying disease risk and explaining the underlying factors contributing to each prediction. The research is expected to contribute a generalized

methodology for **multidisease prediction, explainable clinical AI, multimodal data integration, and continual learning**, thereby providing a foundation for future expansion to additional diseases and healthcare applications.

Proposed 3-paper flow

Paper	Main Focus	Diseases	Algorithm 1	Algorithm 2	Algorithm 3	Learning Type	XAI Methods	Main Experiment	Key Problem	Role in PhD
Paper 1 – Common Framework	Multidisease prediction	DM + CVD + CKD + CLD	LightGBM	CatBoost	TabNet	Supervised learning	SHAP + LIME	Compare 3 algorithms across 4 diseases	Accurate and explainable prediction	Foundation
Paper 2 – Expandable Framework	Continual/expandable learning	DM → CVD → CKD → CLD	Continual LightGBM / Incremental Boosting	Continual Neural Network	Knowledge-Distillation-based Continual Learning	Continual / incremental learning	SHAP + LIME	Add diseases sequentially and measure forgetting	Catastrophic forgetting	Major methodological contribution
Paper 3 – Adaptive & Personalized Framework	Adaptive and personalized prediction	DM + CVD + CKD + CLD + possible new disease	Adaptive LightGBM	Adaptive Deep Neural Network	Multimodal Transformer	Adaptive / continual learning	SHAP + LIME + Counterfactual XAI	Adapt to new/updated patient data	Data adaptation and personalized risk prediction	Final integrated contribution

Simple progression:

Paper 1: Common multimodal + explainable framework

↓

Paper 2: Continual/expandable learning and catastrophic forgetting mitigation

↓

Paper 3: Adaptive + personalized prediction with advanced XAI

Paper 1: An Explainable Multimodal Machine Learning Framework for Early Prediction of Diabetes, Cardiovascular, Chronic Kidney and Chronic Liver Diseases

1. Multimodal Patient Data

Demographic Data
- Age
- Gender
- BMI
- Ethnicity

Clinical Data
- Blood pressure
- Heart rate
- Medical history
- Symptoms
- Lifestyle (smoking, diet, physical activity)

Laboratory Data
- Glucose / HbA1c
- Cholesterol
- Creatinine / eGFR
- Liver enzymes (ALT, AST)
- Other blood parameters

2. Data Preprocessing

Missing value handling → Outlier detection → Encoding (categorical variables) → Normalization / Scaling → Train / Validation / Test split

3. Feature Selection and Multimodal Fusion

Demographic features
Clinical features
Laboratory features
Lifestyle features

Feature Selection
- Correlation analysis
- Mutual information
- Model-based importance

Multimodal Feature Representation

4. Machine Learning Models (Three Algorithms)

LightGBM
- Gradient boosting
- Fast and efficient
- Handles large feature sets

CatBoost
- Categorical variable handling
- Robust to overfitting
- Good for clinical and categorical data

TabNet
- Attention-based neural network for tabular data
- Learns important features
- Handles complex relationships

5. Multidisease Risk Prediction

Diabetes Mellitus (DM)
- Glucose
- HbA1c
- BMI
- Family history
- Other relevant features

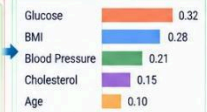
Cardiovascular Disease (CVD)
- Blood pressure
- Cholesterol
- Heart rate
- ECG related features
- Other relevant features

Chronic Kidney Disease (CKD)
- Creatinine
- eGFR
- Blood urea
- Hemoglobin
- Other relevant features

Chronic Liver Disease (CKD)
- ALT, AST
- Bilirubin
- Albumin
- Total protein
- Other relevant features

6. Explainable AI Analysis

SHAP (Global and Local)



- Global feature importance
- Local explanation for each patient

LIME (Patient-level)

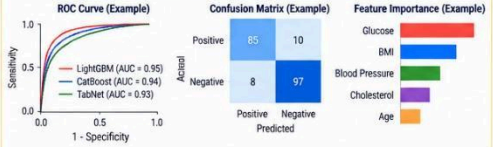
Feature	Contribution
Glucose	+0.28
BMI	+0.20
Blood Pressure	+0.15
Age	+0.10
Cholesterol	−0.88

- Individual patient explanation
- Understand specific prediction

7. Evaluation and Comparison

Disease	Model	Accuracy	Precision	Recall	F1-score	AUROC
Diabetes (DM)	LightGBM	0.92	0.91	0.90	0.91	0.95
	CatBoost	0.91	0.90	0.88	0.89	0.94
	TabNet	0.90	0.89	0.87	0.88	0.93
Cardiovascular (CVD)		...	...	...	...	...
Chronic Kidney (CKD)		...	...	...	...	...
Chronic Liver (CLD)		...	...	...	...	...

8. Results Visualization



9. Key Outcomes of Paper 1

- Common multimodal framework for four diseases
- Comparison of LightGBM, CatBoost and TabNet
- Explainable AI using SHAP and LIME
- Identification of common and disease-specific risk factors
- Accurate and interpretable disease risk prediction

Paper 2: An Expandable Continual Learning Framework for Multidisease Prediction with Catastrophic Forgetting Mitigation

1. Initial Training (Stage 1)

Train with Disease 1: Diabetes (DM)

DM dataset → Base Model (Pretrained on DM) → Output DM risk prediction

Important features (DM)
- Glucose / HbA1c
- BMI
- Age
- Family history
- Other relevant features

2. Sequential Learning of New Diseases

Stage 1 Diabetes (DM)
Train model on DM

Stage 2 + Cardiovascular (CVD)
Incrementally learn CVD (retain DM knowledge)

Stage 3 + Chronic Kidney (CKD)
Incrementally learn CKD (retain DM + CVD knowledge)

Stage 4 + Chronic Liver (CLD)
Incrementally learn CLD (retain DM + CVD + CKD knowledge)

3. Continual Learning Strategies

Incremental LightGBM
- Online/incremental update
- Fast and efficient
- Handles new disease data

Continual Neural Network (DNN)
- Regularization
- Replay mechanism
- Learns new disease without forgetting

Knowledge Distillation
Teacher Model (previous knowledge) → Distillation → Student Model (updated model)
- Transfers old knowledge
- Reduces forgetting
- Maintains performance on previous diseases

4. Model Training Process

DM data
CVD data
CKD data
CLD data

Continual Learning Framework
- Sequential training
- Knowledge retention
- Forgetting mitigation
- Model update

Trained Model
- Predicts all 4 diseases
- Retains previous knowledge

5. Multidisease Prediction (Final Model)

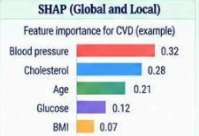
Input: New Patient Data
(Demographic + Clinical + Laboratory + Lifestyle)

Continual Learning Model (Updated)

Diabetes (DM)	Cardiovascular (CVD)	Chronic Kidney (CKD)	Chronic Liver (CLD)
Risk: 65%	Risk: 72%	Risk: 28%	Risk: 40%

6. Explainable AI Analysis

SHAP (Global and Local)



LIME (Patient-level)

Patient X – CVD risk = 72%

Feature	Contribution
Blood pressure	+0.25
Cholesterol	+0.20
Age	+0.15
Glucose	+0.10
BMI	−0.05

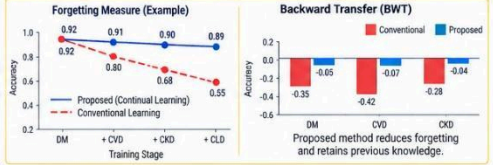
Explanation Stability



7. Performance Evaluation

Disease	Model	Accuracy	Precision	Recall	F1-score	AUROC
Diabetes (DM)	Conventional	0.90	0.88	0.87	0.88	0.94
	Continual (Proposed)	0.91	0.90	0.89	0.90	0.95
Cardiovascular (CVD)	Conventional	0.88	0.85	0.84	0.84	0.92
	Continual (Proposed)	0.90	0.88	0.87	0.88	0.94
Chronic Kidney (CKD)	Conventional	0.86	0.83	0.82	0.82	0.90
	Continual (Proposed)	0.89	0.87	0.86	0.86	0.93
Chronic Liver (CLD)	Conventional	0.83	0.80	0.78	0.79	0.88
	Continual (Proposed)	0.87	0.84	0.83	0.83	0.91

8. Catastrophic Forgetting Analysis



9. Key Outcomes of Paper 2

- Model can learn new diseases sequentially
- Reduces catastrophic forgetting
- Retains knowledge of previous diseases
- Maintains stable explanations (SHAP/LIME)
- Accurate multidisease prediction
- Expandable framework for additional diseases
- Suitable for real-world clinical use

Paper 3 in very simple words

Paper 1:
The model learns **4 diseases**.

Paper 2:
The model learns **new diseases without forgetting the old diseases**.

Paper 3:
The model learns from **new patient data and adjusts itself**, while explaining **why each patient received a particular risk prediction**.

Simple example

Suppose your system has already learned:

DM + CVD + CKD + CLD

A new patient comes into the system.

Patient information

- Age: 55
- BMI: high
- Blood pressure: high
- Glucose: high
- Creatinine: high
- eGFR: low

The model processes this information.

Step 1 — Prediction

It might produce:

Disease	Model risk
Diabetes	82%
CVD	68%
CKD	89%
CLD	25%

These are **model-generated risk estimates**, not medical diagnoses.

Step 2 — Explanation

The system then says, for example:

CKD risk = 89%

Main contributing factors:

- High creatinine
- Low eGFR
- High blood pressure
- High glucose
- Age

This is the **Explainable AI** part.

Step 3 — New patients arrive

After some time, you receive many new patients whose characteristics are somewhat different from the original training data.

Instead of completely rebuilding the model from the beginning:

Old model

⬇️

New patient data

⬇️

Adaptive learning

⬇️

Updated model

The model learns from the new information while trying to **retain its previous knowledge**.

Step 4 — Personalized explanation

Two patients could both have:

CKD risk = 80%

But their reasons may be different.

Patient A:

High creatinine + low eGFR are the major contributors.

Patient B:

High blood pressure + diabetes-related features are the major contributors.

So the model doesn't just give:

“80% risk.”

It also gives:

“These are the factors that influenced this patient's prediction.”

Where the algorithms come in

For Paper 3, you can compare:

Adaptive LightGBM
↓
Adaptive DNN
↓
Multimodal Transformer

Then use:

SHAP + LIME + Counterfactual XAI

Suppose you divide your data into different time/population groups.

Phase 1

Train the model using:

Dataset A

↓

Evaluate.

Phase 2

Introduce:

Dataset B

↓

Adapt the model.

Phase 3

Introduce:

Dataset C

↓

Adapt again.

Then evaluate:

1. Performance on new data
2. Performance on old data
3. Knowledge retention
4. Computational cost

5. Explanation stability

This can make Paper 3 stronger.

For example:

Training population

Age: 30–60

New population

Age: 50–80

Or the distribution of laboratory values changes.

You investigate:

Does the model's performance degrade when the data distribution changes?

Then:

Can adaptive learning recover/improve performance?

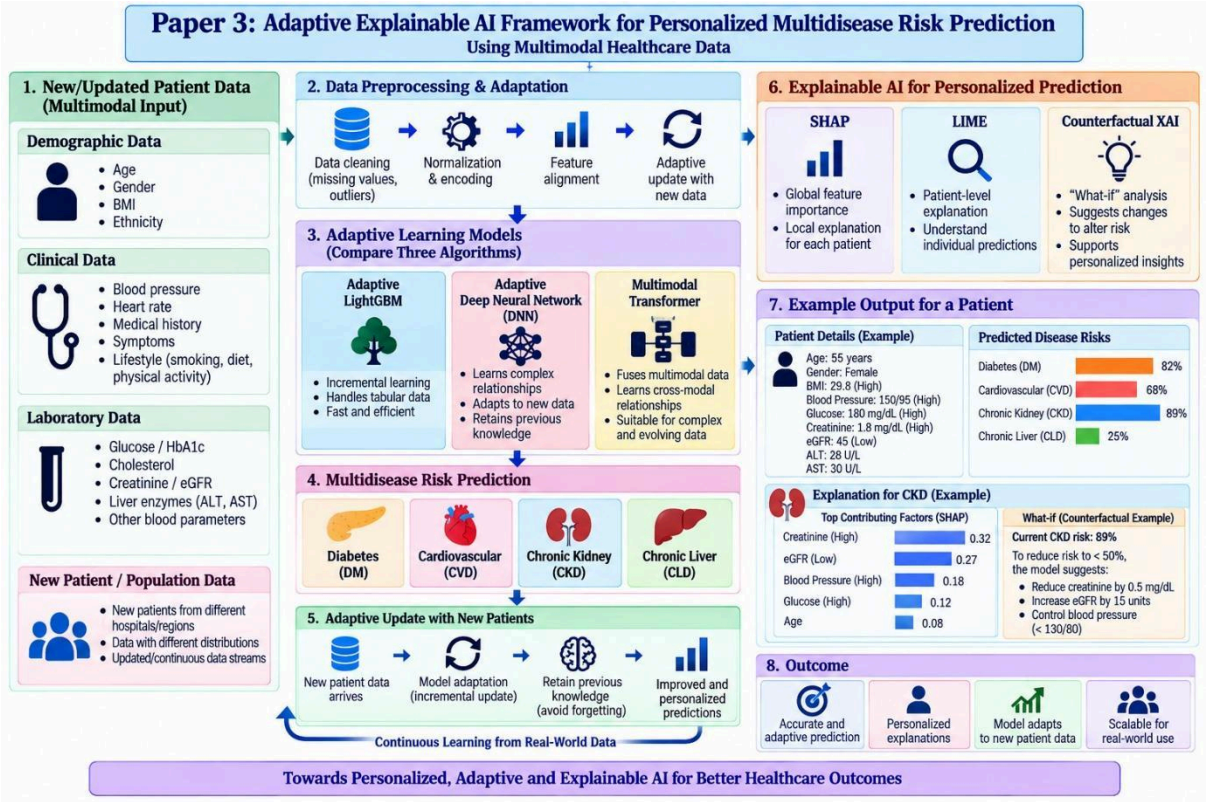
This gives Paper 3 a clear research problem.

So the simplest definition of Paper 3 is:

“Develop an AI system that can adapt to new patient data and provide personalized explanations for multidisease risk predictions.”

The PhD story becomes:

PAPER 1 → PREDICT
PAPER 2 → EXPAND
PAPER 3 → ADAPT & PERSONALIZE



S.NO.	Work	Results	Paper Writing	Publishcations	Total
1	Paper1	30k	25k	-	55,000
2	Paper2	35k	25k	60000	1,20,000
3	Paper3	40k	25k	60000	1,25,000
					3,00,000

Thesis Writing:

1.Index

2.Introduction

3.Literature Survey

4.Paper1

5.Paper2

6.Paper3

7.Comparision of all 3 papers

8.Conclusion

9.Biobilography-200 references papers

10.Final viva presentation

11.Synposis and per synopsis document +ppt

Total cost = 50,000

S.No.	Work	Cost
1	Paper pubilcations	3,00,000
2	Thesis	50,000
	Total	3,50,000

Step-by-step research workflow for your journal paper

Target journal: Network Modeling and Analysis in Health Informatics and Bioinformatics

Proposed title: An Explainable Multimodal Machine Learning Framework for Early Prediction of Diabetes, Cardiovascular, Chronic Kidney and Chronic Liver Diseases

We can design the paper using just three existing algorithms—LightGBM, CatBoost and Random Forest—and one proposed framework. The study should be planned so that every experiment supports a research question, and every table and figure provides evidence for your contribution.

One important clarification: no workflow or set of tables can guarantee acceptance. The aim is to build a rigorous, reproducible paper that fits the journal's network-modelling focus.

Overall workflow

Step 1. Select four public disease datasets

Diabetes · Cardiovascular · CKD · Liver disease

Step 2. Data cleaning and preprocessing

Missing values · Encoding · Class imbalance · Leakage prevention

Step 3. Train three baseline algorithms

LightGBM · CatBoost · Random Forest

Step 4. Develop the proposed framework

Feature-association network + prediction + explainability

Step 5. Evaluate all four approaches

Performance · Network properties · Explainability · Ablation

Step 6. Statistical validation and discussion

Confidence intervals · Robustness · Limitations · Reproducibility

Step 1. Finalize the research contribution

Before coding, define exactly what your proposed algorithm adds beyond the three baselines.

I recommend a Network-Assisted Explainable Disease Prediction Framework with these components:

1. Feature-association network construction.
2. Network analysis to identify central features and disease-related feature groups.
3. Prediction using a clearly defined model that uses network-derived information.
4. SHAP-based explanations.
5. Evaluation against LightGBM, CatBoost and Random Forest.

The journal emphasizes new methodologies for network-based modelling, knowledge discovery and health decision support. Therefore, the network component must contribute technically to the method, rather than simply appear as a visualization.



PJIP

+1

Step 2. Select datasets and document them

Choose one suitable public dataset per disease. Use a consistent evaluation protocol, but preprocess each dataset according to its own clinical variables.

Table 1 — Dataset characteristics

Disease	**Required information**
Diabetes	Dataset source, sample count, class distribution and features
Cardiovascular disease	Dataset source, sample count, class distribution and features
Chronic kidney disease	Dataset source, sample count, class distribution and features
Chronic liver disease	Dataset source, sample count, class distribution and features

Figure 1 — Research workflow: dataset selection → preprocessing → network construction → four model configurations → explainability → validation.

Important: if the four datasets contain different patients and different features, they do not automatically form a multimodal dataset. Unless you have genuinely linked modalities, consider revising the title to “An Explainable Network-Based Machine Learning Framework for Multi-Disease Prediction.”

Step 3. Build the three baseline models

Train LightGBM, CatBoost and Random Forest separately for each disease.

Table 2 — Baseline model performance

Disease	Model	Accuracy	Sensitivity	Specificity	F1-score	AUROC	AUPRC
Diabetes	LightGBM	—	—	—	—	—	—
Diabetes	CatBoost	—	—	—	—	—	—
Diabetes	Random Forest	—	—	—	—	—	—
Repeat for other 3 diseases							

Calculate these metrics from actual test predictions. Report confidence intervals and repeated-run variability where appropriate. Avoid selecting thresholds using the test set.

Step 4. Construct the proposed network

Build a separate feature-association network for each disease, using a justified method to define nodes and edges.

Table 3 — Network characteristics

Disease	Nodes	Edges	Density	Top central features	Communities
Diabetes	—	—	—	—	—
Cardiovascular disease	—	—	—	—	—
Chronic kidney disease	—	—	—	—	—
Chronic liver disease	—	—	—	—	—

Calculate degree and betweenness centrality, community structure and network stability where appropriate.

Figure 2 — Disease-specific networks: visualize the nodes, edges and communities. Include a legend explaining node size, node colour and edge strength.

Figure 3 — Centrality comparison: compare the most central features for the four diseases.

Be careful not to interpret statistical associations as proof of biological causation.

Step 5. Implement the proposed algorithm

The proposed method should use information from the network to improve prediction or provide a demonstrable analytical benefit.

For example, compare:

- A baseline classifier using the original features.

- The same classifier augmented with network-derived features.
- Your complete proposed framework.

Table 4 — Ablation study

Configuration	Original features	Network-derived features	SHAP explanations	AUROC	F1-score
Baseline	Yes	No	Post-hoc	—	—
Proposed variant A	Yes	Yes	Post-hoc	—	—
Proposed variant B	Yes	Yes	Post-hoc	—	—
Full proposed method	Yes	Yes	Yes	—	—

Define each variant precisely in the Methods section. SHAP is an explanation technique, not automatically a predictive component, so its presence alone does not establish a new algorithm.

Step 6. Generate explainability results

Table 5 — Top explanatory features

Disease	Feature	Mean absolute SHAP value	Direction of effect	Stability across runs
Diabetes	—	—	—	—
Cardiovascular disease	—	—	—	—
Chronic kidney disease	—	—	—	—
Chronic liver disease	—	—	—	—

Prepare:

- Figure 4: SHAP summary plots for each disease.
- Figure 5: Local explanations for selected test predictions.
- Figure 6: Comparison of feature importance across diseases.

Discuss whether the network identifies important features that are also influential in the predictive models. Do not claim clinical significance without supporting evidence.

Step 7. Perform statistical validation

Table 6 — Robustness and statistical analysis

Test	Result to report
Repeated evaluation	Mean and standard deviation

Confidence intervals	95% intervals for primary metrics
Model comparisons	Appropriate paired statistical tests
Missing-data robustness	Performance at predefined missingness levels
Calibration	Brier score and calibration plots
External validation	Performance on independent data, if available
Computational cost	Training time, inference time and model size

Figure 7: ROC and precision–recall curves.

Figure 8: Calibration curves.

Figure 9: Robustness comparison across experimental conditions.

Ensure that all preprocessing and feature selection are fitted on training data only. If you use resampling, perform it inside the training folds.

Step 8. Prepare the final results section

A practical manuscript structure is:

Results subsection	**Main evidence**
4.1 Dataset characteristics	Table 1
4.2 Baseline performance	Table 2
4.3 Network analysis	Table 3, Figures 2–3
4.4 Proposed method and ablation	Table 4
4.5 Explainability	Table 5, Figures 4–6
4.6 Statistical validation and robustness	Table 6, Figures 7–9

You can combine figures or tables where appropriate to avoid making the paper unnecessarily long.

Final recommendation

The most important results for this journal are:

1. A clearly novel network-based method.
2. Fair comparison against the three selected baseline algorithms.
3. Evidence that the network component contributes useful information.
4. Disease-wise explainability.

5. Robustness, statistical validation and reproducibility.

Do not start by generating tables with assumed accuracy values. First select the datasets, implement the method and calculate the results.

Your next step: choose the four datasets. Once those are fixed, the network design, feature selection and experiment protocol can be specified precisely.